

Modeling High-Risk Booking Loss in Hotel Reservations

A Classification and Time-Series Analysis

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Business Problem & Objective

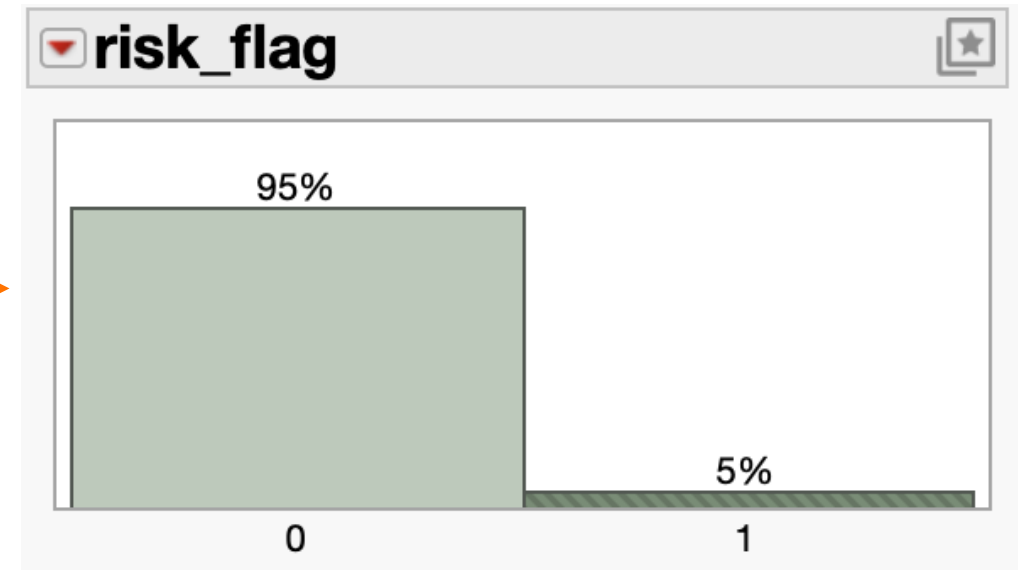
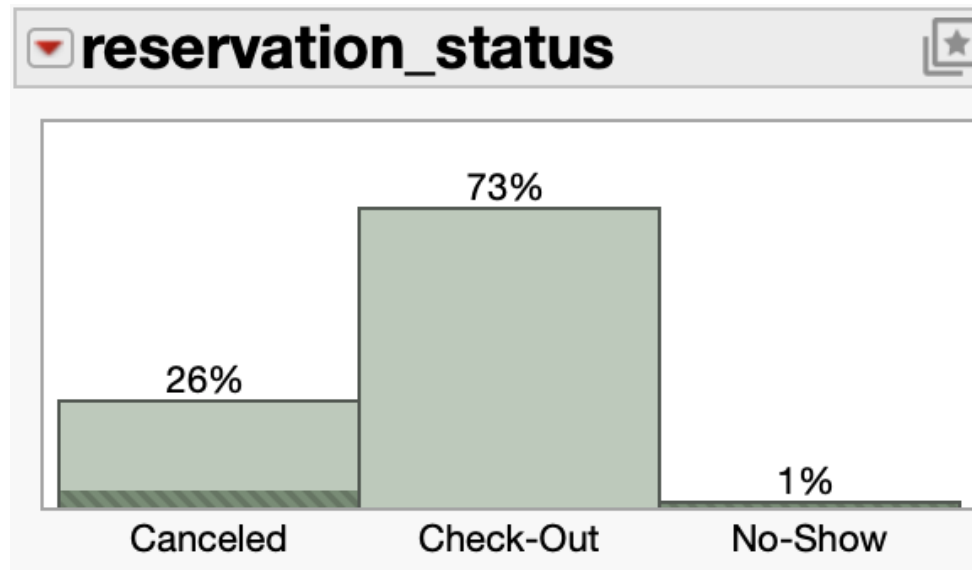
Problem

- Not all cancellations are equally costly
- **Late cancellations** and **no-shows** create financial loss

Objective

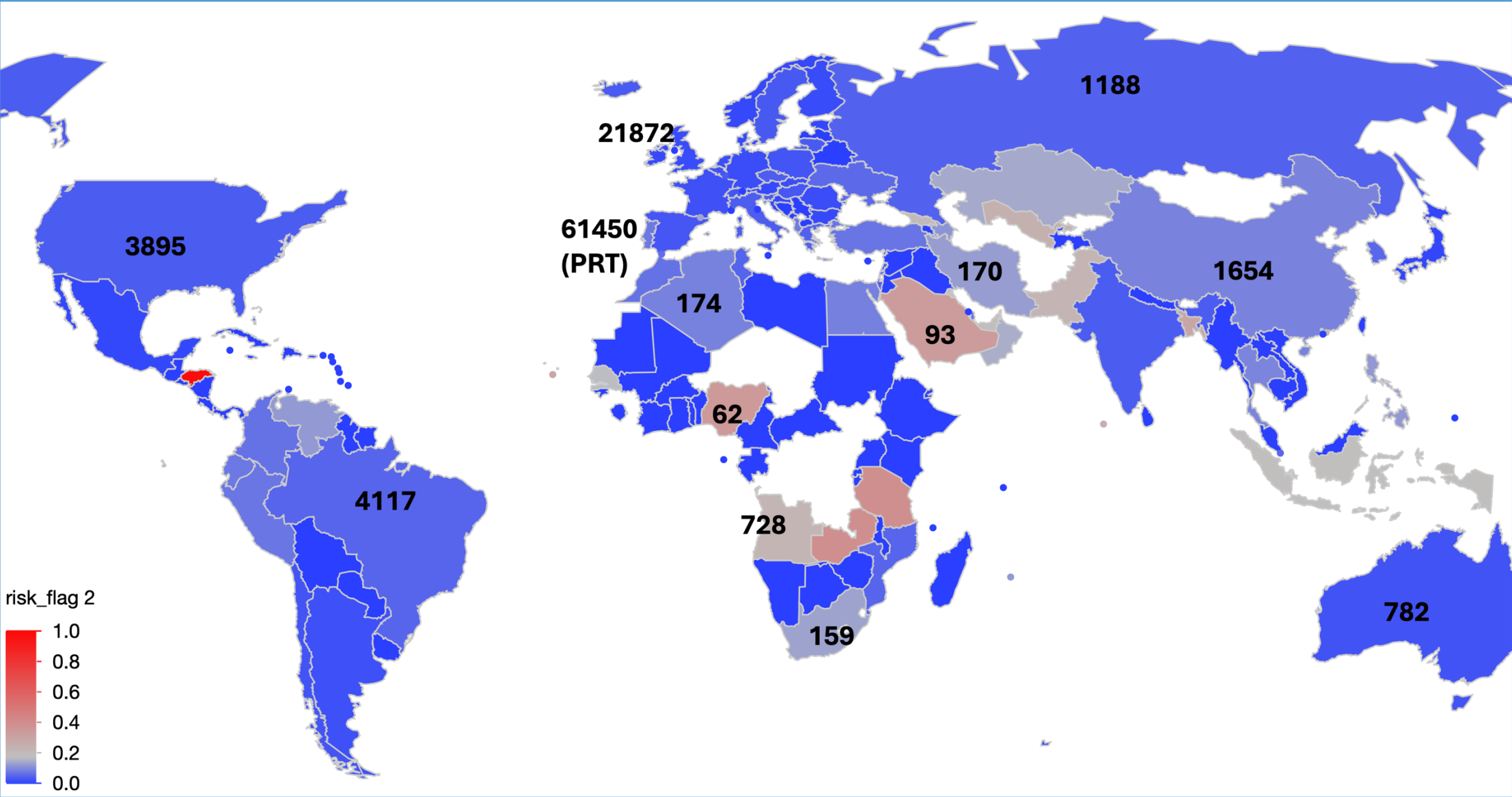
- Identify high-risk bookings
- Support better overbooking decisions

Defining High-Risk Bookings

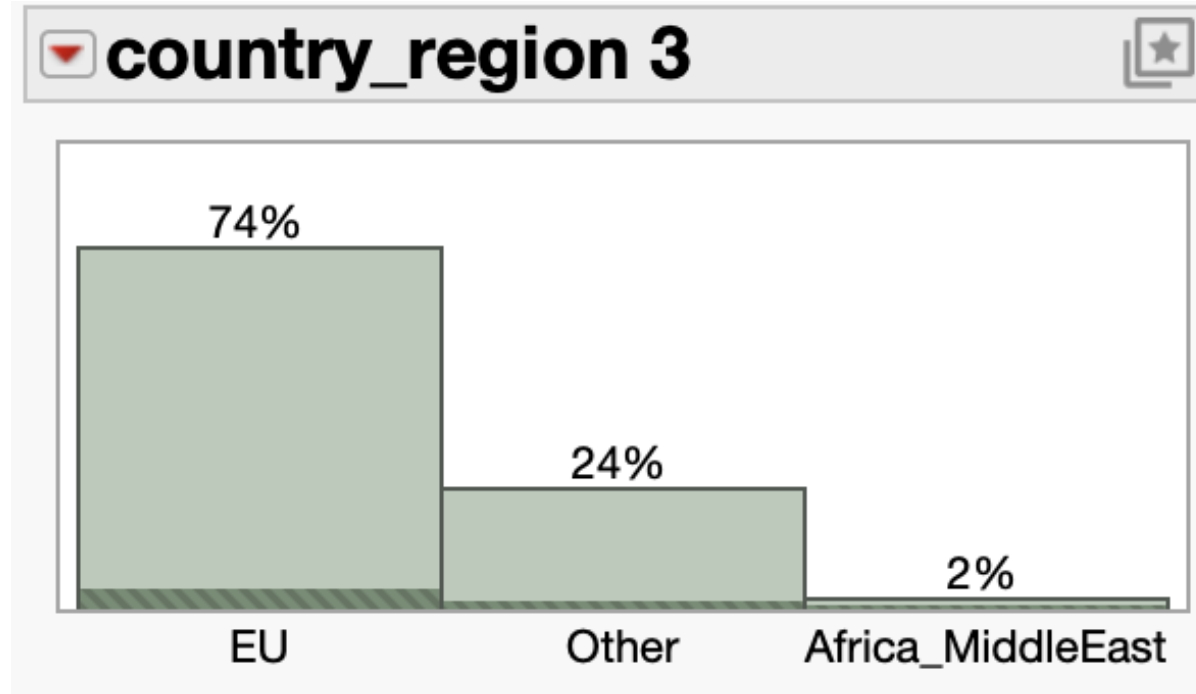


! High-risk cases were assigned higher weight (20:1) to address class imbalance

Geographic Patterns in Bookings and Risk



Customer Distribution by Region



*** High-risk bookings are observed across all regions**

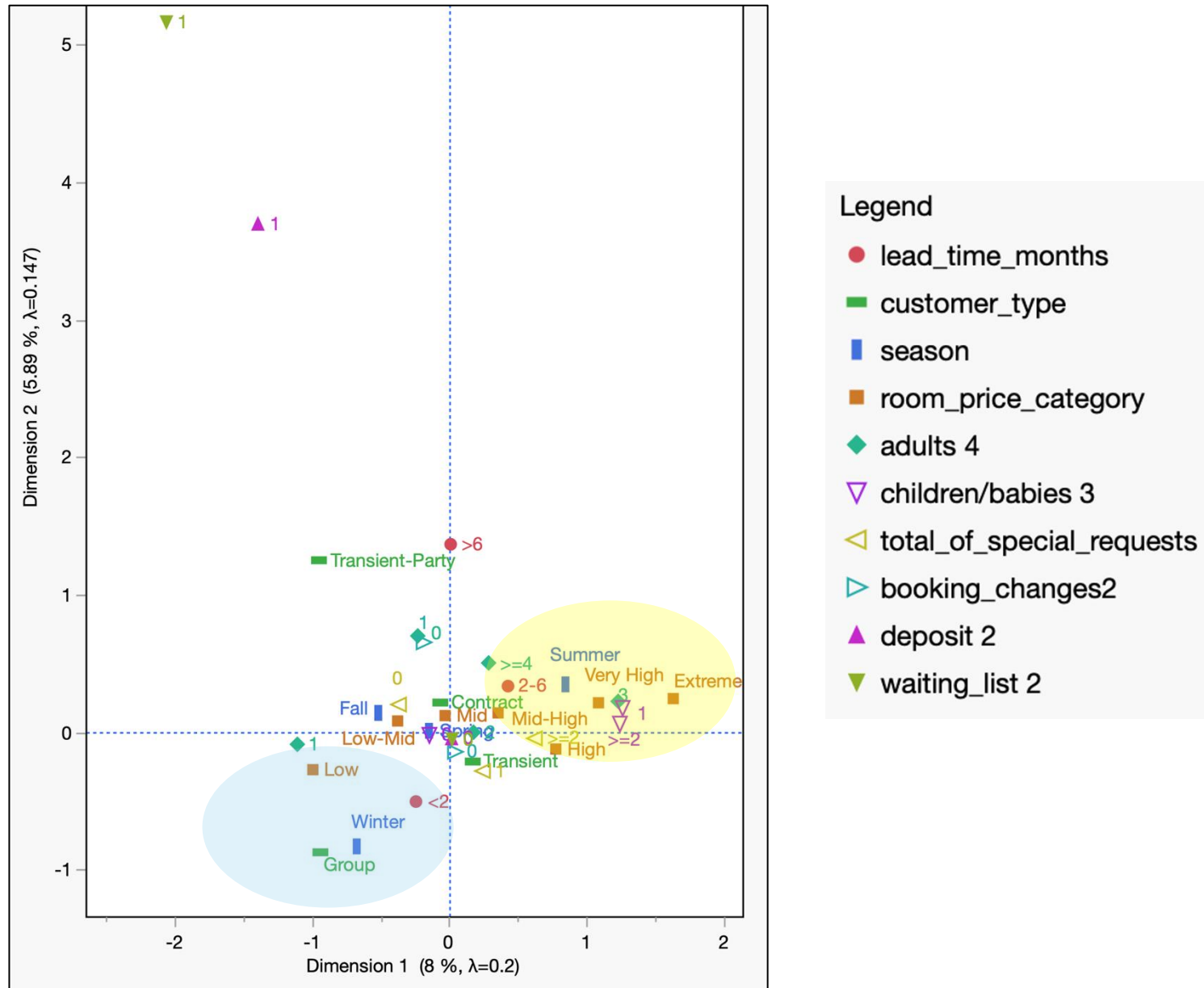
Room Price Categorization



Further Feature Preparation



Exploring Booking Patterns: MCA

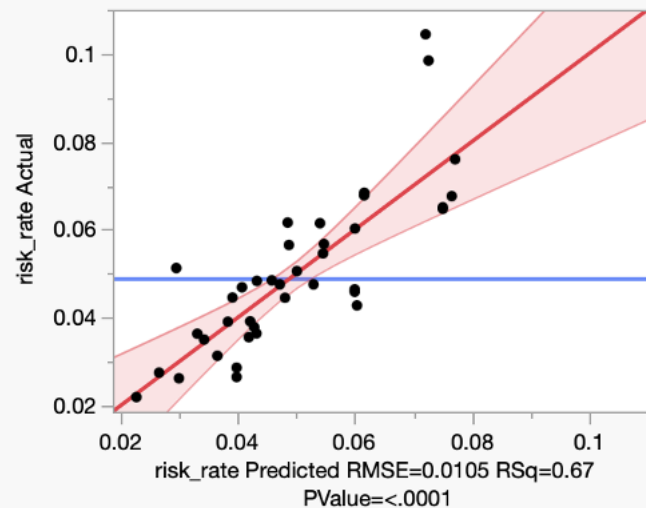


Exploring Booking Patterns: LCA

Cluster	Overall	customer_type	lead_time_months	season	room_price_category
Planned Seasonal Travelers	0.46884				
Short-Lead Budget Travelers	0.39529				
Group / Event Travelers	0.13587				

Exploratory Regression of Monthly Risk

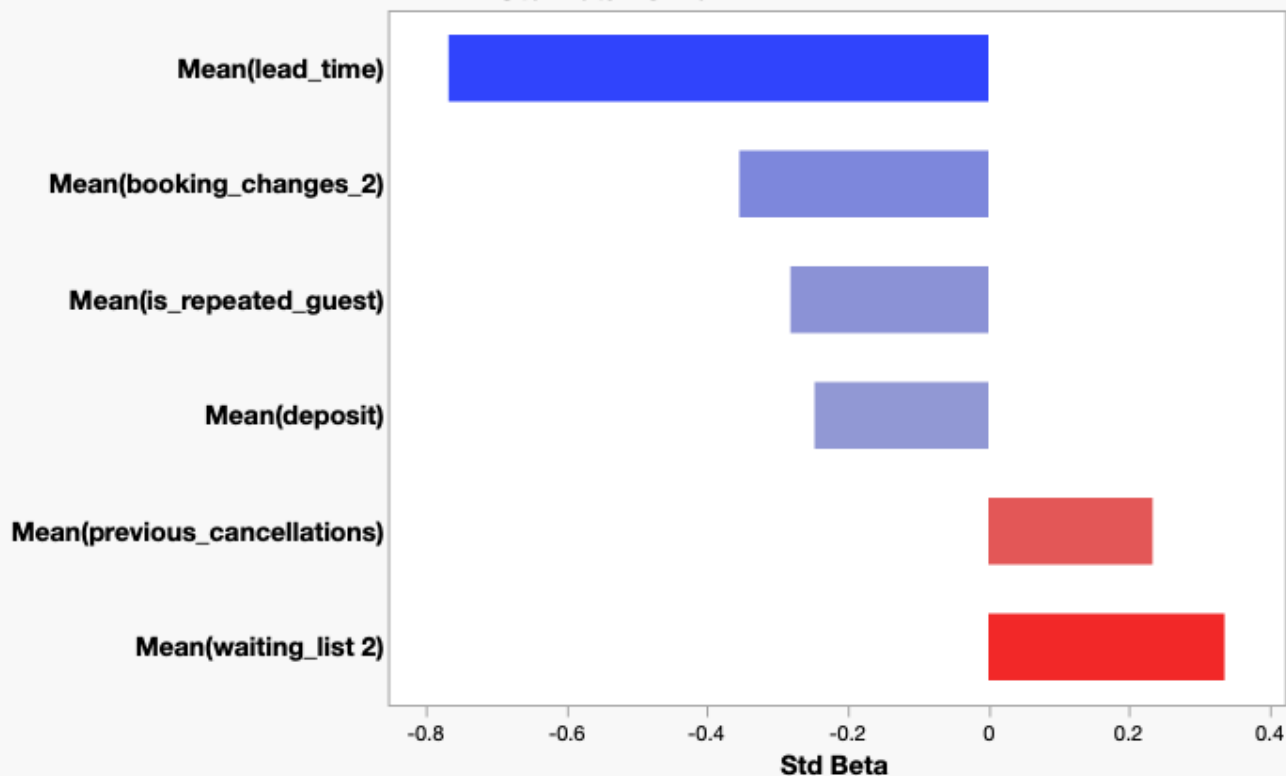
Actual by Predicted Plot



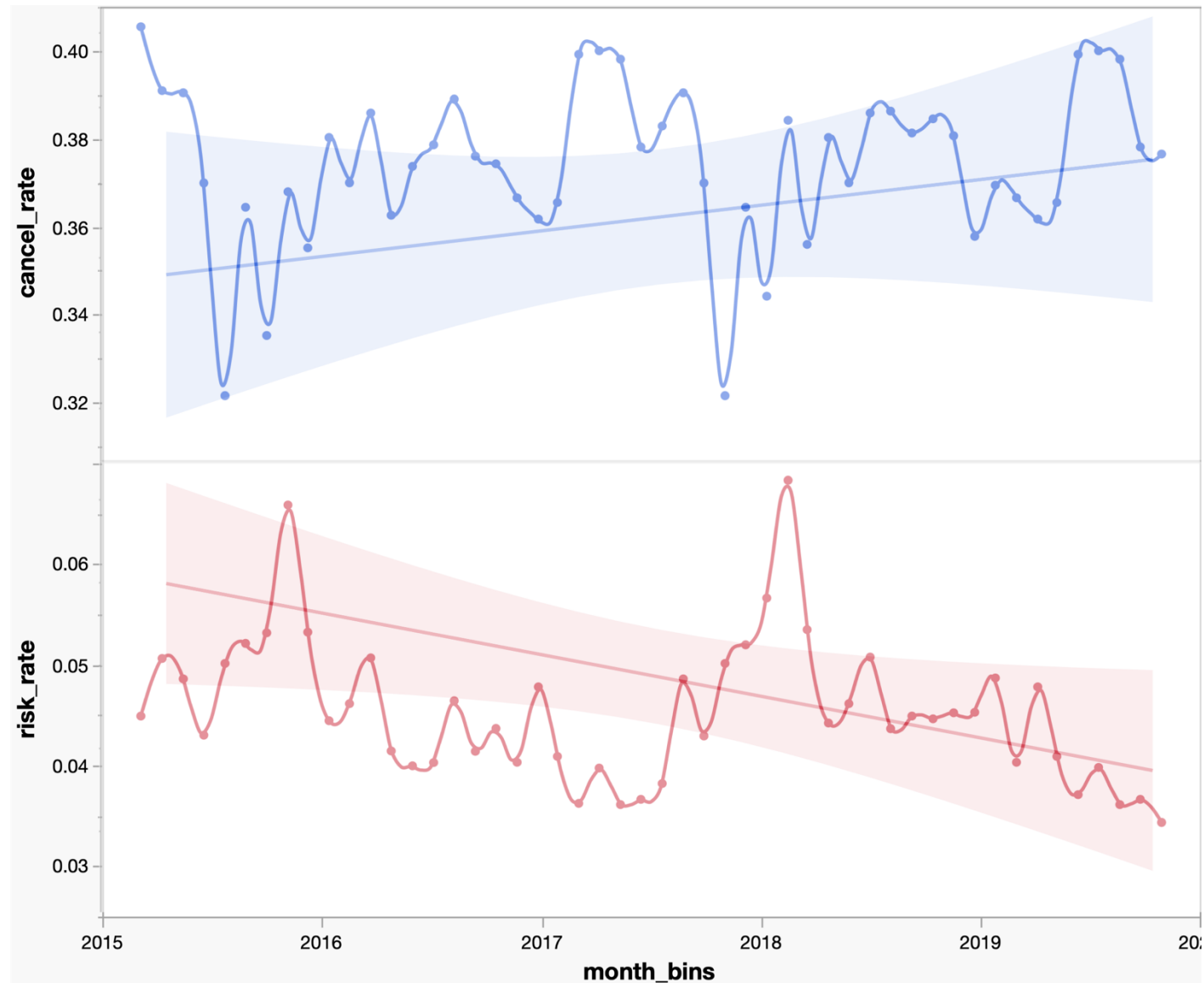
Effect Summary

Source	Logworth	PValue
Mean(lead_time)	4.539	0.00003
Mean(waiting_list 2)	2.671	0.00213
Mean(booking_changes_2)	1.654	0.02218
Mean(previous_cancellations)	1.565	0.02721
Mean(is_repeated_guest)	1.112	0.07732
Mean(deposit)	0.993	0.10165

Std Beta vs. Term



Temporal Patterns in Cancellations and Risk



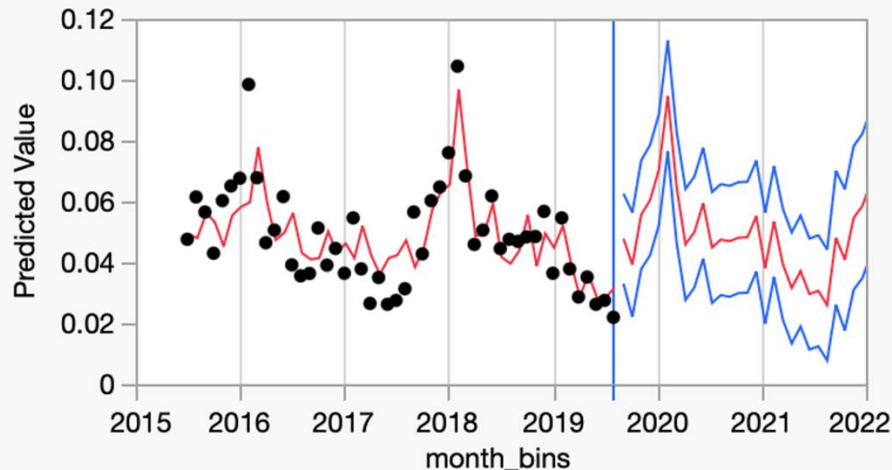
Seasonal ARIMA as a Time Structure Diagnostic

Model: Seasonal ARIMA(1, 0, 0)(1, 0, 0)24

Parameter Estimates

Term	Factor	Lag	Estimate	Std Error	t Ratio	Prob> t	Constant	Mu
AR1,1	1	1	0.58507459	0.1207909	4.84	<.0001*	Estimate	0.04958332
AR2,24	2	24	0.83806650	0.0592710	14.14	<.0001*	0.00333152	
Intercept	1	0	0.04958332	0.0062717	7.91	<.0001*		

Forecast



RSquare 0.60724523
RSquare Adj 0.59053226
MAPE 18.3588538
MAE 0.00812983

- Strong lag-1 dependence
- 24-month seasonal structure confirmed
- $R^2 \approx 0.60$ (consistent with exploratory model)

Assessing Lagged Effects of External Factors

Transfer Function Model (1)

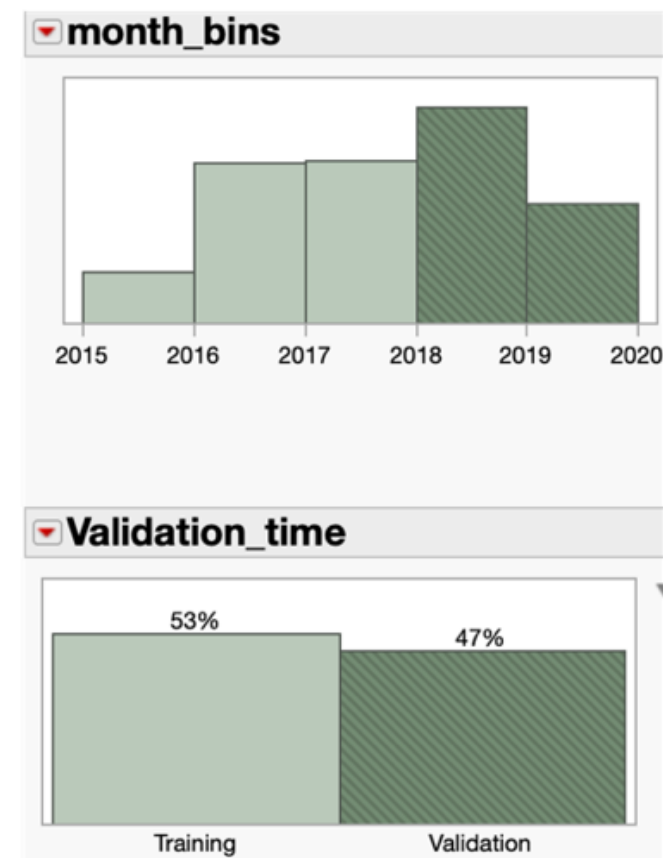
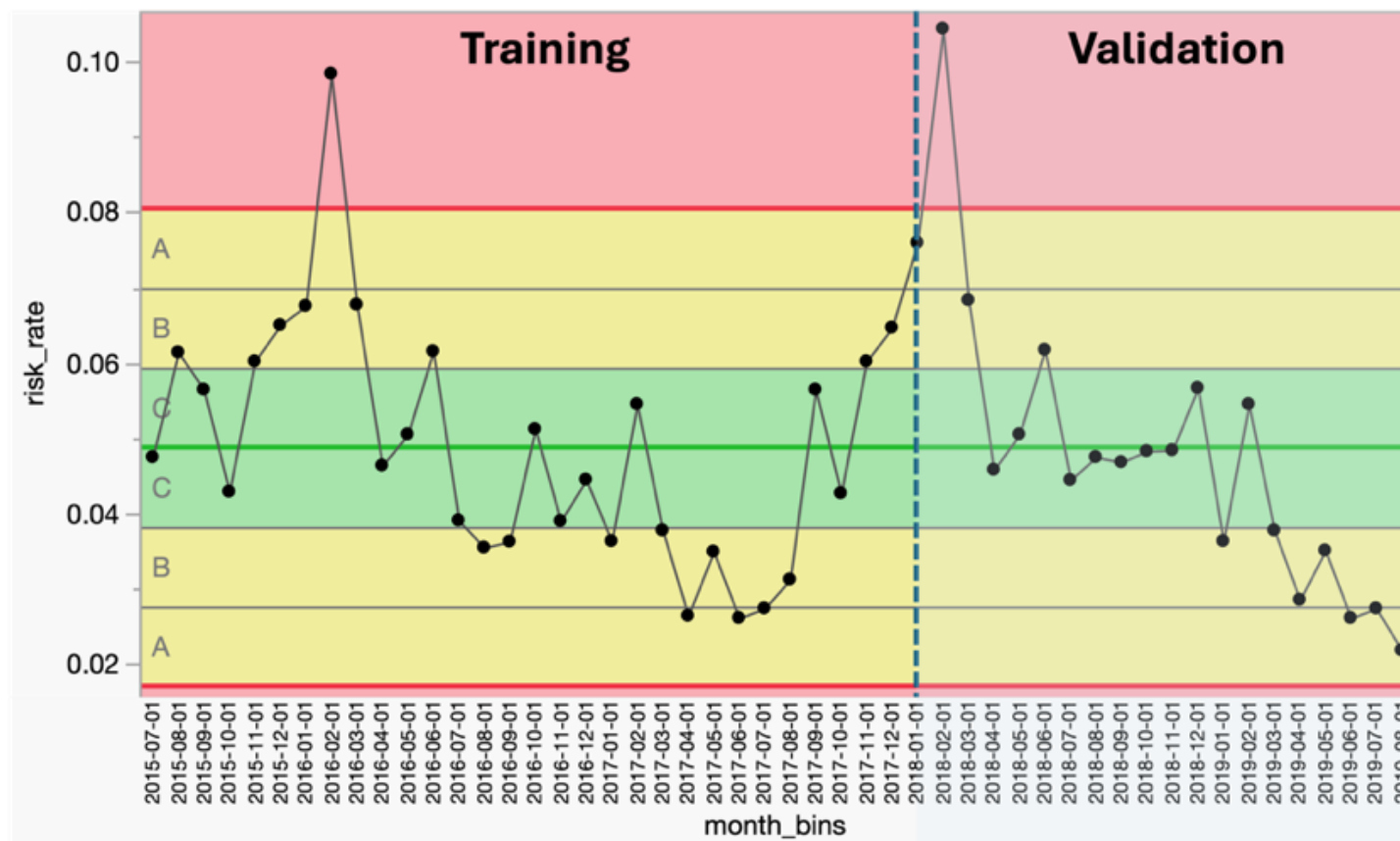
Model Summary

DF	41
Sum of Squared Innovations	0.00523519
Sum of Squared Residuals	0.00523519
Variance Estimate	0.00012769
Standard Deviation	0.0112999
Akaike's 'A' Information Criterion	-277.08276
Schwarz's Bayesian Criterion	-267.93955
RSquare	0.63548496
RSquare Adj	0.60308362
MAPE	14.6377451
MAE	0.00683845
-2LogLikelihood	-287.08276

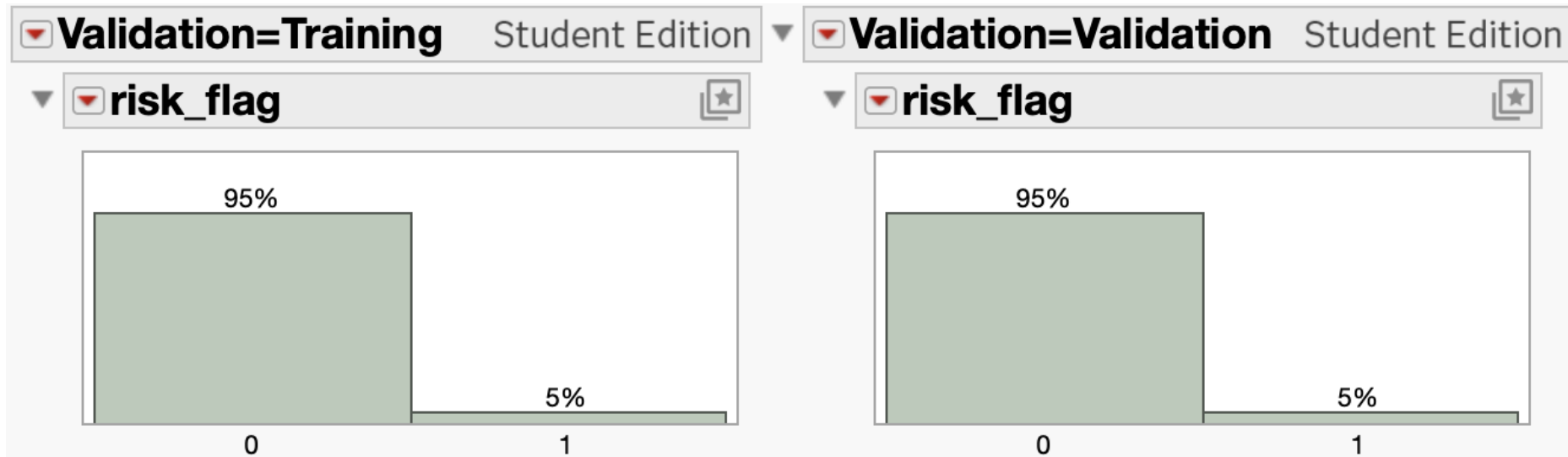
Parameter Estimates

Variable	Term	Factor	Lag	Estimate	Std Error	t Ratio	Prob> t
	Intercept	0	0	0.1087410	0.0144585	7.52	<.0001*
Mean(adr)	Num0,0	0	0	-0.0004022	0.0000850	-4.73	<.0001*
Mean(total_of_special_requests)	Num0,0	0	0	-0.0829175	0.0195392	-4.24	0.0001*
Mean(booking_changes_2)	Num0,0	0	0	0.1326091	0.0594529	2.23	0.0312*
risk_rate	AR1,1	1	1	0.3093153	0.1390261	2.22	0.0317*

From Time-Series Insights to Classification



Validation Design: Class Balance



Model Screening: Selecting Candidate Models

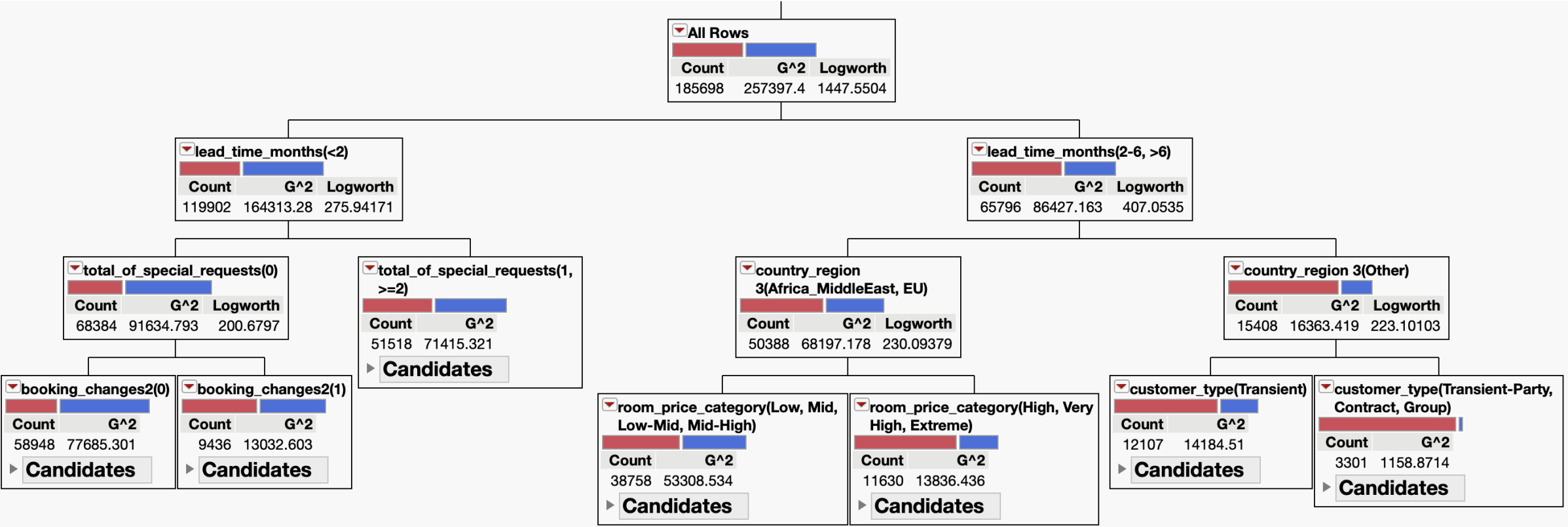
Model Screening for risk_flag

Student Edition

Validation

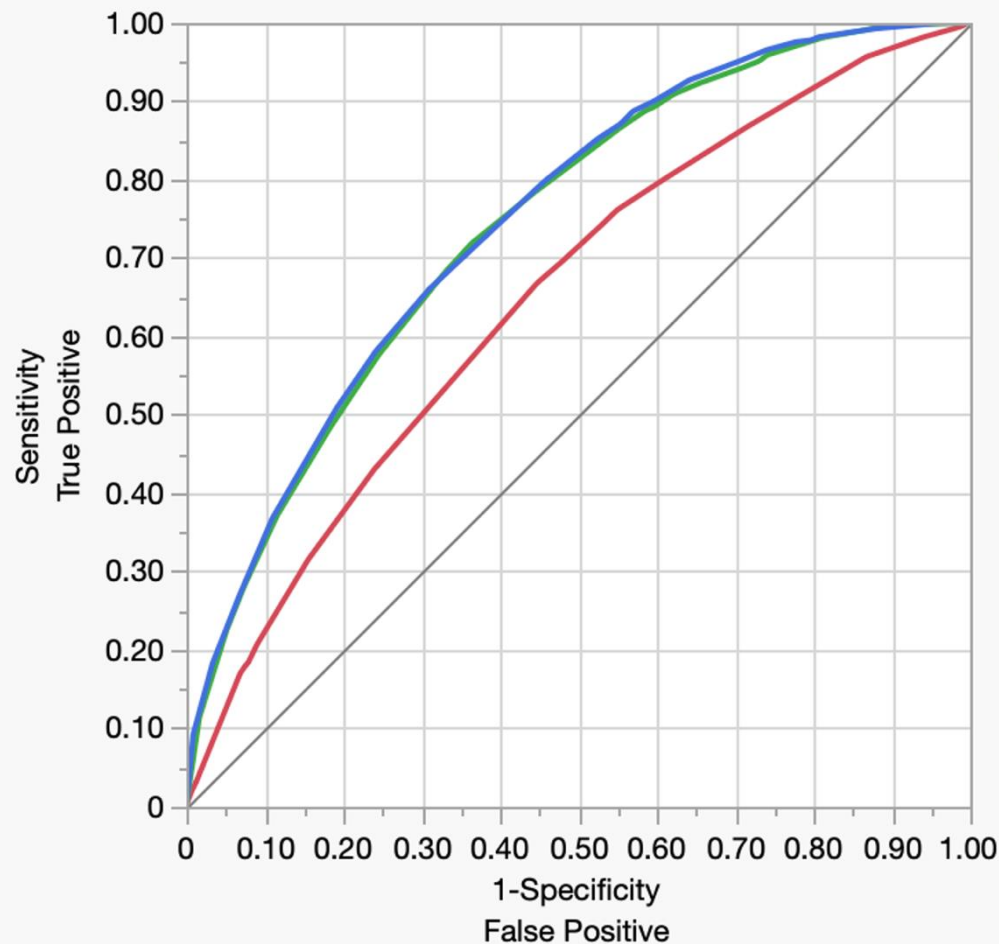
Method		N	Sum Freq	Sum Wgt	Entropy RSquare	Misclassification Rate	AUC	RASE	Generalized RSquare
Boosted Tree	†	88830	88830	165989	0.1546	0.3248	0.7490	0.32104	0.2572
Bootstrap Forest	†	88830	88830	165989	0.1434	0.3274	0.7458	0.45294	0.2404
Decision Tree	†	88830	88830	165989	0.0894	0.3695	0.6859	0.47095	0.1554
Generalized Regression Lasso	†	88830	165989	165989	0.0606	0.3835	0.6624	0.47927	0.1073
Fit Stepwise	†	88830	88830	165989	0.0605	0.3832	0.6624	0.47927	0.1073
Nominal Logistic		88830	88830	165989	0.0605	0.3834	0.6624	0.47928	0.1073

High-Risk Booking Profiles (Decision Tree)



Comparing Model Performance (Validation Set)

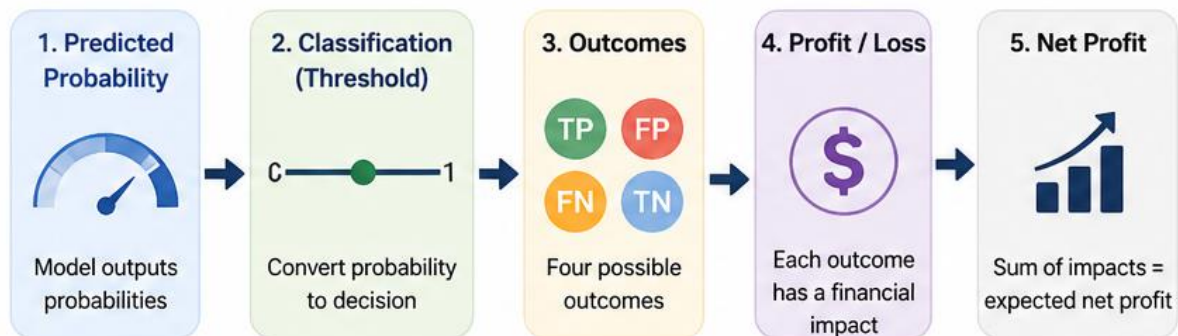
ROC Curve for risk_flag=1



Predictor	Creator	AUC
DT_Prob(risk_flag==1)	Partition	0.6490
BootstrapForest_Prob(risk_flag==1)	Bootstrap Forest	0.7441
BoostedTree_Prob(risk_flag==1)	Boosted Tree	0.7490

Business Optimization: Choosing the Right Threshold

From Prediction to Profit



Profit Matrix (Economic Impact of Outcomes)

Outcome	Definition	Interpretation	Profit / Loss
True Positive (TP)	Predicted high risk & actually high risk	Recoverable booking value through overbooking	+ M
False Positive (FP)	Predicted high risk but actually low risk	Unnecessary overbooking (costly action)	- 1.5 × M
False Negative (FN)	Predicted low risk but actually high risk	Missed loss-prevention opportunity	- M
True Negative (TN)	Predicted low risk & actually low risk	Business as usual (baseline)	0

M = 70% of room revenue (proxy for contribution margin)

Net Benefit Across Different Thresholds

Threshold	Sum(Profit)	Relative Ranking
0.30	-26,940,495.13	8
0.40	-1,798,781.80	7
0.50	24,316,353.82	6
0.60	48,167,626.85	5
0.70	60,399,381.57	4
0.80	64,759,047.53	3
0.90	65,145,926.45	★ 1
0.91	65,120,101.39	2
0.92	65,100,535.00	3
0.95	65,064,788.05	4



Optimal Threshold ≈ 0.90

This threshold maximizes expected net benefit.

Although sensitivity is low (~2.5%), it minimizes costly false positives and delivers the highest incremental business value under the assumed cost structure.

Key Insights and Business Implications

1 Risk has temporal structure

- Short-term persistence (lag 1)
- Seasonal recurrence (24-month pattern)

2 Risk is multi-factor, not single-driver

- Combination of booking behavior and customer profile
- No single predictor is sufficient on its own

3 Model performance is moderate

- Boosted Tree performs best
- Discrimination remains limited

4 Business value comes from decision strategy



- Profit matrix enables decision optimization
- Optimal threshold $\approx$ **0.90**
- High-confidence predictions
→ positive net benefit

Thank you!

